

Flexible Job Shop Scheduling with Job Precedence Constraints: A Deep Reinforcement Learning Approach (Supplementary File)

Yishi Li, Chunlong Yu

APPENDIX A DETAILED SCHEDULING PROCESS

This section provides details on the scheduling process. Fig. A1 illustrates the step-by-step scheduling procedure in a heterogeneous disjunctive graph for a 4-job, 3-machine instance. The first five steps are explained as follows:

- 1) Initially, the system clock is $T = 0$, and three operations are available for scheduling: $O_{1,1}$, $O_{2,1}$, and $O_{3,1}$. In our approach, an operation becomes active for scheduling when all its precedence operations have either been completed or have started processing.
- 2) Suppose the selected arc at this step is $\{O_{2,1}, M_1\}$, meaning that $O_{2,1}$ is scheduled on machine M_1 . Since M_1 is idle at this moment, the start time is $st = 0$, and the completion time is calculated as $ct = 7$, given that the processing time of $O_{2,1}$ on M_1 is 7. The active operation set is then updated: since $O_{2,1}$ has started at $T = 0$, $O_{2,2}$ is now available for scheduling. The active operation set becomes $\{O_{1,1}, O_{2,2}, O_{3,1}\}$.
- 3) Suppose the selected arc at this step is $\{O_{3,1}, M_3\}$. The start and completion times are calculated as 0 and 9, respectively. The active operation set updates to $\{O_{1,1}, O_{2,2}, O_{3,2}\}$.
- 4) Suppose the selected arc at this step is $\{O_{3,2}, M_2\}$. Since $O_{3,2}$ can only start after the completion of $O_{2,1}$ at $T = 7$, its start time is 7, and its completion time is 11. The active operation set updates to $\{O_{1,1}, O_{3,2}\}$.
- 5) At this point, all machines connected to the active nodes are busy. The system clock is advanced to the earliest time when an operation is completed, i.e., $T = 7$.

This process continues until all operations are completed.

APPENDIX B DETAILED FEATURES IN THE GRAPH

In this section, details for the two types of node features and the undirected arc features are provided.

1) Operation features:

- (o1) Scheduling flag: 0 if O_{ij} is unscheduled otherwise 1.
- (o2) Estimated completion Time: defined by Equation (20) of the paper.
- (o3) Minimum processing time among all machines.
- (o4) Span of processing time among all machines.
- (o5) Average processing time among all machines.
- (o6) Waiting time: the time from the ready time of O_{ij} until now t .

- (o7) Remaining processing time: the time from now t until the estimated completion Time (0 for unscheduled O_{ij}).
- (o8) Number of remaining operations: the number of unscheduled operations from O_{ij} to End along the directed arc.
- (o9) Remaining work: the sum of average processing time of unscheduled operations from O_{ij} to End along the directed arc.

- (o10) Number of machines that O_{ij} can be processed on.

2) Machine features:

- (m1) Number of candidate operations that M_k can process.
- (m2) Number of unscheduled operations that M_k can process.
- (m3) Minimum processing time of operations that M_k can process.
- (m4) Average processing time of operations that M_k can process.
- (m5) Waiting time: The time elapsed since the machine became free until the current time T .
- (m6) Remaining processing time: The time from the current time T until the machine becomes free (0 if M_k is free).
- (m7) Free time: the moment when M_k is free.
- (m8) Working flag: 0 if M_k is free otherwise 1.

3) Undirected arc features:

- (a1) Processing time p_{ijk} .
- (a2) Ratio of p_{ijk} to the maximum processing time of O_{ij} .
- (a3) Ratio of p_{ijk} to the maximum processing time of candidate operations that M_k can process at current time.
- (a4) Ratio of p_{ijk} to the maximum processing time of unscheduled operations.
- (a5) Ratio of p_{ijk} to the maximum processing time of unscheduled operations that M_k can process.
- (a6) Ratio of p_{ijk} to the maximum processing time of all feasible (O_{ij}, M_k) .
- (a7) Ratio of p_{ijk} to the remaining work of O_{ij} .
- (a8) Summation of the waiting time of O_{ij} and M_k .

Here, compared to those proposed in Wang et al. [12], the newly designed features include (o2), (o8), (o9), and (a3). These features are specifically adapted to the tree-like Job Precedence Constraint (JPC) in our problem. More specifically, when computing the estimated completion time of operation O_{ij} in (o2), we consider all operations related to O_{ij} through JPC across the entire graph. Instead of assuming $C_{i0} = 0$, we determine the estimated starting time of C_{i0} by Equation (20). Similarly, when calculating the number of remaining operations (o8) and the remaining work (o9), we account for all operations connected to O_{ij} via JPC, i.e., all

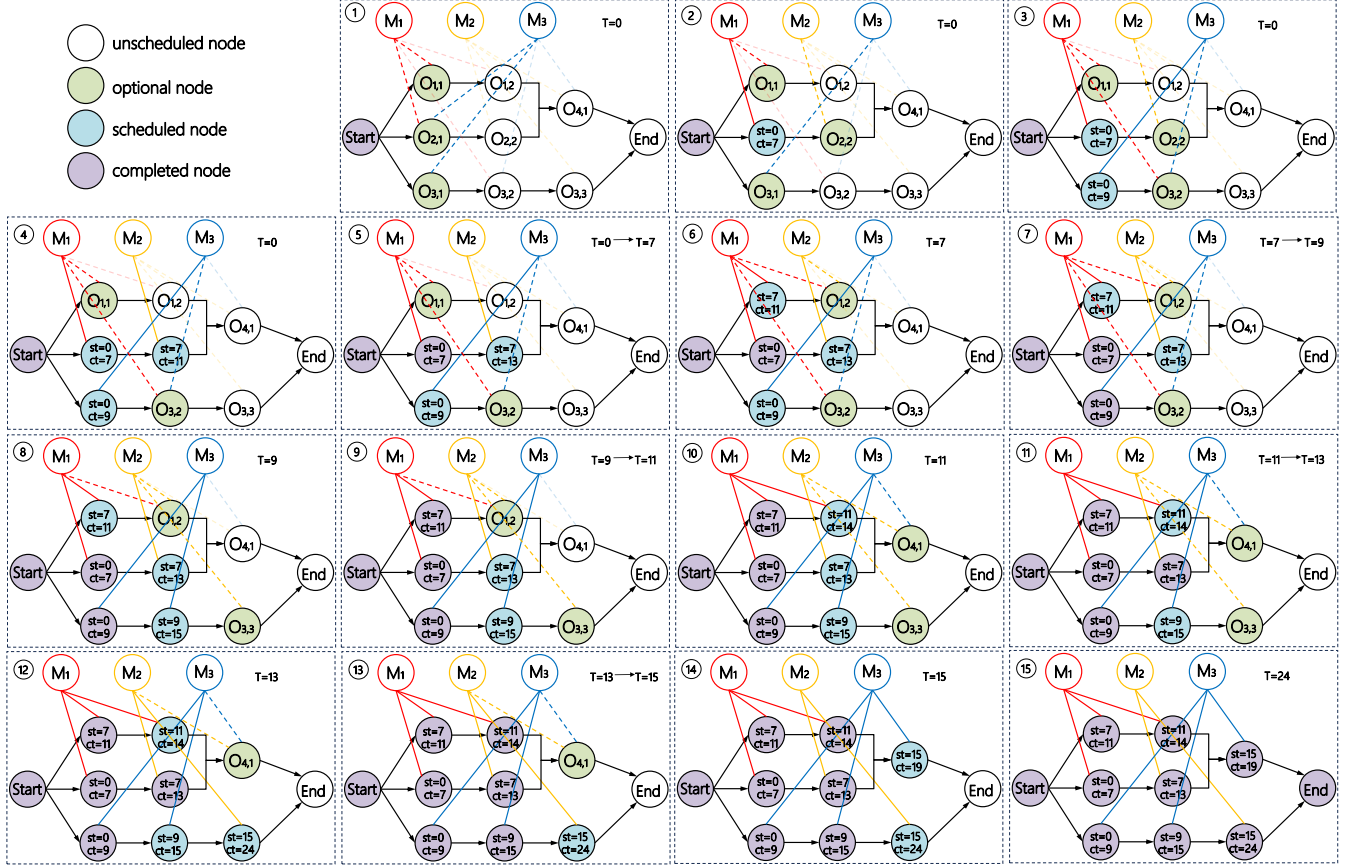


Fig. A1: An example of the scheduling process

unscheduled operations along the directed path from O_{ij} to End . On the other hand, the calculation of feature (a3) differs from that in Wang et al. [12]. Since we use a different action space, the set of candidate operations that a machine, say M_k , can process at a given time T is larger than in the original version.

APPENDIX C DETAILS OF THE DATA GENERATION

In this section, details for the result of the data generation are provided.

We define that each assembly job group with job precedence constraints having the following characteristics:

- 1) There are 12 operations within 6 jobs, where the number of operations is sampled without replacement from a set $\{1, 1, 2, 2, 3, 3\}$.
- 2) There are three manufacturing stages, as the example Fig. 1. $\{J_1, J_2, J_5\}$ is the first stage with no predecessor jobs, $\{J_3\}$ is the second stage with both predecessor and successor jobs, and $\{J_6\}$ is the third stage, the final assembly stage, can begin only after all other jobs have been completed.
- 3) Referring to the data in Zhu and Zhou [4], the processing time for each operation is sampled from $U(100, 200)$.
- 4) All operations in the final assembly stage can only be assigned to a specific machine for processing. While

for each remaining operation, two feasible machines are sampled from a remaining machine list without repetition.

APPENDIX D DETAILED RULES OF COMPARISON

In this section, details for PDR rules are provided.

- 1) First In First Out (FIFO): Select the candidate operation that is ready at the earliest time and assign it to the first available eligible machine.
- 2) Most Operations Remaining (MOR): Select the candidate operation that has the highest number of remaining successor operations (cross-job), and assign it to a random machine that is immediately available.
- 3) Shortest Processing Time (SPT): Select the (O_{ij}, M_k) combination with the shortest processing time.
- 4) Most Work Remaining (MWKR): Select the candidate operation with the highest average remaining processing time of remaining successor operations (cross-job), along with a random machine that is immediately available to process the operation.

APPENDIX E DETAILED RESULTS OF DESIGN OF EXPERIMENTS

In this section, we present detailed results from the design of experiments. Fig. A2 shows the boxplot of the average relative gap for MGPPPO. Here, OR1–OR8 represent the factor

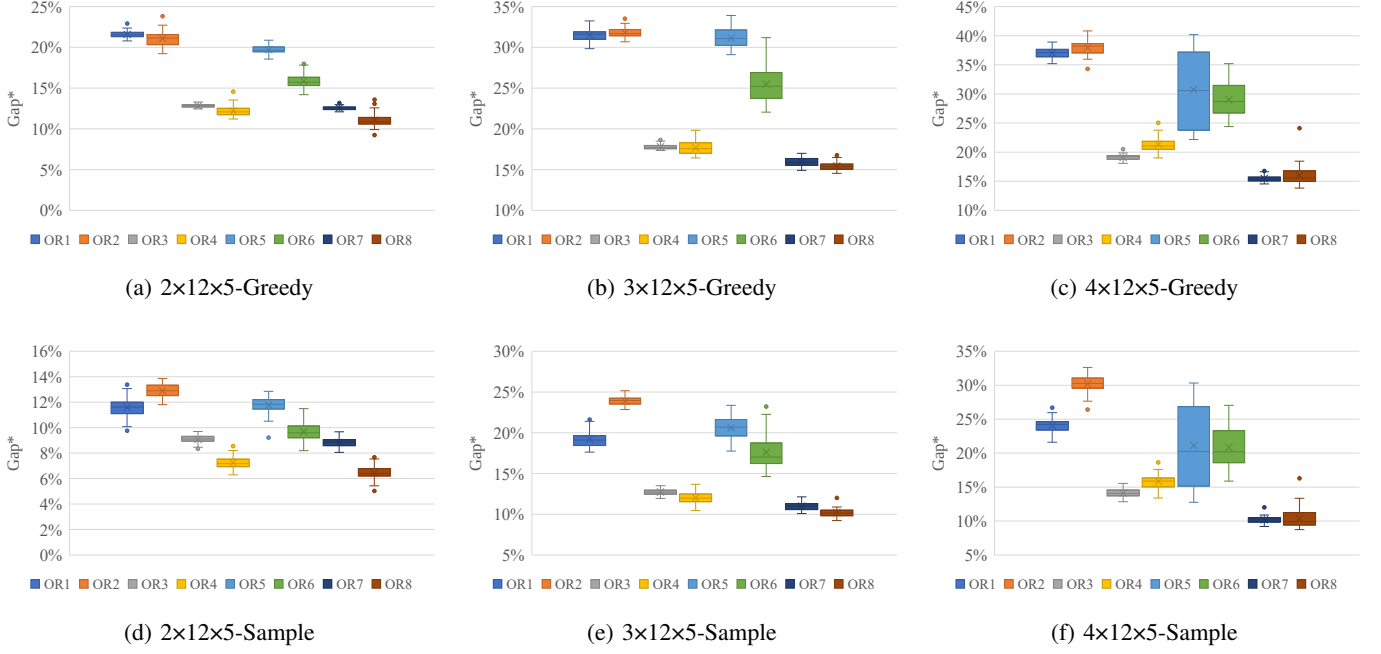


Fig. A2: Performance of factor combinations OR1 to OR8.

TABLE I: Notations for factor combinations

Exp. ID	Information	Node feature	Action space
OR1	Single-job	Old	No waiting
OR2	Single-job	Old	Allow waiting
OR3	Single-job	New	No waiting
OR4	Single-job	New	Allow waiting
OR5	Cross-job	Old	No waiting
OR6	Cross-job	Old	Allow waiting
OR7	Cross-job	New	No waiting
OR8	Cross-job	New	Allow waiting

combinations listed in Table I, with each boxplot summarizing results from solving 1,000 instances. Fig. A2 (a)–(c) correspond to the Greedy strategy, while Fig. A2 (d)–(f) correspond to the Sampling strategy. OR1 represents the baseline setting without any modifications to the mechanisms, whereas OR8 corresponds to the configuration proposed in this paper. As shown, OR8 outperforms all other settings across nearly all problem sizes and strategies.